Today, after Quiz 2, we explained why sufficiency is important. Let's recap quickly.

Sufficiency is used to capture a property about data sets and their summaries, which has to do with compressing without losing information. In a Bernoulli setup, $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} \text{Bern}(\theta)$ (i.e., $P_\theta(X_i = x) = \theta^x(1-\theta)^{1-x}$ for $x \in \{0, 1\}$), we saw that

$$\text{lik}(\theta) = \prod_{i=1}^{n} \theta^x (1-\theta)^{1-x} = \left(\frac{\theta}{1-\theta}\right)^{\sum_{i=1}^n X_i} (1-\theta)^n.$$

Maximizing $\text{lik}(\theta)$ over θ produces the MLE,

$$\hat{\theta}_{\text{MLE}} = \frac{1}{n}\sum_{i=1}^{n} X_i,$$

which is the proportion of 1's in the data.

> ☞ **Observation 1.** If we didn't have the full data $\{X_i\}_{i=1}^n$ but only the summary statistic $T(X_1, \ldots, X_n) = \sum_{i=1}^n X_i$, we could still obtain the MLE by computing T over n.

Observation 1 suggests that when we wish to use a data set to infer parameters of interest, we might not actually need the entire data set, but perhaps only a *compressed summary* of it. But how do we know if a choice of compressed summary works? Sufficiency lays down a concrete criterion for this.

Sufficiency

Definition 1 (Sufficiency) *A statistic $T = T(X_1, \ldots, X_n)$ is* sufficient *for $p_\theta(x)$ if, given its value, say $T = t$, the distribution of the data $X_1, \ldots, X_n$ does not depend on the parameter θ. That is, $P_\theta(X_1, \ldots, X_n \mid T = t)$ doesn't depend on θ.*

We saw in section that for Bernoulli-distributed X_i's, given $T(X_1, \ldots, X_n) = \sum_{i=1}^n X_i$ and a binary sequence of length n given by $(x_1, \ldots, x_n) \in \{0, 1\}^n$,

$$P_\theta(X_1 = x_1, \ldots, X_n = x_n \mid T = t) = 1 \Big/ \binom{n}{t},$$

which doesn't depend on θ. So T is sufficient for the Bernoulli family parametrised by θ.

> ☞ **Observation 2.**
>
> 1. The entire data set, $\{X_i\}_{i=1}^n$, is sufficient. This is because, if T is the function that maps the data set to itself, then
>
> $$P_\theta(X = x | T = t) = \begin{cases} 1 & \text{if } x = t \\ 0 & \text{otherwise} \end{cases}.$$

2. In the Bernoulli example with $n = 100$, $T = (X_1 + \ldots + X_{99}, X_2 + \ldots + X_{100})$ (so T is a vector of length 2 with first component capturing the sum of odd-indexed observations and second component capturing the sum of even-indexed observations) is sufficient. Why? Use the same reasoning as we did in section to show that $P_\theta(X_1 = x_1, \ldots, X_n = x_n \mid T = t)$ doesn't depend on θ for this specific function T. (Hint: the random variables are i.i.d..)

3. Definition 1 looks like a bad working definition. If we're asked to prove that something is sufficient, are we supposed to compute the conditional distribution and show that it doesn't depend on θ? Isn't that kinda not easy already for the Bernoulli example??? Ugh!! ***waves hands angrily***

Observation 2 makes three important points. The first shows that, even without actually compressing our data set, we still have information to construct and compute the MLE. This makes sense, it's just what we've been seeing in class prior to talking about sufficiency. (Take derivative of lik(θ), set to zero, yada yada.)

(Skip the second point first.)

The third point motivates the following question: Is there a simpler way to check if a statistic is sufficient? The answer is yes, thanks to the following result due to Ronald Fisher and Jerzy Neyman.

Theorem 2 (Neyman-Fisher Factorisation, p. 306 of Rice) *Given data $X = (X_1, \ldots, X_n)$ with $X_1, \ldots, X_n \overset{i.i.d.}{\sim} p_\theta(x)$, $T = T(X_1, \ldots, X_n)$ is sufficient for θ (or for p_θ)* if and only if *there exist non-negative functions g_θ and h satisfying*

$$p_\theta(X) = g_\theta(T)h(X). \tag{1}$$

Let's apply Theorem 2 to some examples. First look at the Bernoulli example. Since

$$p_\theta(X) = \left(\frac{\theta}{1-\theta}\right)^{\sum_{i=1}^n X_i} (1-\theta)^n, \tag{2}$$

if we let $n = 100$ and consider $T = \sum_{i=1}^{100} X_i$, then the functions $g_\theta(t) = \left(\frac{\theta}{1-\theta}\right)^t (1-\theta)^n$ and $h(x) = 1$ provide a valid factorisation of p_θ, showing that T is sufficient for the Bernoulli distribution parametrised by θ. Next, considering $S = (S_1, S_2) = (X_1 + \ldots + X_{99}, X_2 + \ldots + X_{100})$ (a two-dimensional compressed summary of the data, like in the second point of Observation 2), the functions $g_\theta(s) = \left(\frac{\theta}{1-\theta}\right)^{s_1+s_2} (1-\theta)^n$ and $h(x) = 1$ provide a valid factorisation of p_θ. So S is also sufficient.

Next look at a Poisson example. If $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} \text{Pois}(\theta)$, then $p_\theta(x) = e^{-\theta}\theta^x/x!$ for $x \in \mathbb{N}$, so

$$p_\theta(X) = e^{-n\theta}\theta^{\sum_{i=1}^n X_i} \Big/ (X_1! \cdots X_n!).$$

Again, letting $T(X) = \sum_{i=1}^n X_i$, the functions $g_\theta(t) = e^{-n\theta}\theta^t$ and $h(x) = 1/(x_1! \cdots x_n!)$ provide a valid factorisation of p_θ. So T is sufficient for the Poisson distribution parametrised by θ. If we let

$S(X) = (X_1 + \ldots + X_{99}, X_2 + \ldots + X_{100})$ as in the Bernoulli example, it's not difficult to construct g_θ and h and to show that S is also sufficient.

These examples lead nicely to the second point of Observation 2. So many things are sufficient: one-dimensional summaries (e.g., the sum), two-dimensional summaries (e.g., the sum of odd-indexed and the sum of even-indexed), and even the data itself! We need a criterion to tell us if a sufficient statistic is good or not!

Minimal Sufficiency

Minimal sufficiency is THE criterion. We saw, in the Bernoulli example, how T and S and the data set itself are all sufficient. However, it's clear that T and S are lower-dimensional summaries of the data set, and moreover T itself is a summary of S. Perhaps the criterion should look out for whether one sufficient statistic can be summarised from another. Bingo!

Definition 3 (Minimal Sufficiency) *A statistic $T = T(X_1, \ldots, X_n)$ is* minimal sufficient *for $p_\theta(x)$ if for any other sufficient statistic $S = S(X_1, \ldots, X_n)$, T is a function of S. That is, there exists a map ("compression") f such that $T = f(S)$.*

Let's see if this definition gels with what we've seen so far. Yes, we see that $T = S_1 + S_2$, i.e., the map $f(S) = S_1 + S_2$ maps S to T. So T is more compressed than S.

☞ **Observation 3.** Since T is one-dimensional, and you can't compress beyond a scalar-valued summary of the data, T's got to be minimal sufficient! But how are we going to verify this using the definition of minimal sufficiency? Are we expected to check that all sufficient statistics can be compressed to T??? Holy moly!! ***stares emotionlessly into space***

Observation 3 is a fair response. Thankfully, like for sufficiency, we have a good working definition for minimal sufficiency (i.e., not the actual definition but one that can be implemented in practice).

Theorem 4 (Minimal Sufficiency "Working Definition" / Criterion) *Let $T = T(X)$ be a summary of the data set $X = (X_1, \ldots, X_n)$. If, for any two data sets X and Y,*

$$\frac{p_\theta(X)}{p_\theta(Y)} \text{ doesn't depend on } \theta \iff T(X) = T(Y), \tag{3}$$

then T is minimal sufficient.

Let's apply Theorem 4 to the Bernoulli example. Let's consider each of S and T, to see why one is minimal sufficient and the other is not. Let's first look at T. If $T(X) = T(Y)$, then since by definition $T(x) = \sum_{i=1}^n x_i$, eq. (2) implies that

$$\frac{p_\theta(X)}{p_\theta(Y)} = \left(\frac{\theta}{1-\theta}\right)^{\sum_{i=1}^n X_i} (1-\theta)^n \Big/ \left(\frac{\theta}{1-\theta}\right)^{\sum_{i=1}^n Y_i} (1-\theta)^n = 1,$$

so the ratio of likelihoods doesn't depend on θ. This verifies the $\Leftarrow$ direction. To verify the $\Rightarrow$ direction, we check the *contrapositive*: suppose $T(X) \neq T(Y)$, then $\frac{p_\theta(X)}{p_\theta(Y)}$ depends on θ. This is easy to check: if $T(X) \neq T(Y)$, then $\frac{p_\theta(X)}{p_\theta(Y)} = [\theta/(1-\theta)]^{T(X)-T(Y)}$ depends on θ. (For example, if

$T(X) = 31$ and $T(Y) = 42$ then this ratio is $[\theta/(1-\theta)]^{-11}$, which is a function of θ.) Thus T is indeed minimal sufficient, by the working definition.

What about S? If $S(X) = S(Y)$ then it's easy to verify that $\frac{p_\theta(X)}{p_\theta(Y)} = 1$, so $\Leftarrow$ direction holds. However, if $S(X) \neq S(Y)$, is it true that $\frac{p_\theta(X)}{p_\theta(Y)}$ depends on θ? Consider $S(X) = (11, 10)$ and $S(Y) = (10, 11)$, i.e., the odd-indexed observations returned 11 heads for data set X and the even-indexed observations returned 10 heads for the data set X (etc.). Since $S_1(X) + S_2(X) = 11 + 10 = 10 + 11 = S_1(Y) + S_2(Y)$, and $S_1(x) + S_2(x) = \sum_{i=1}^n x_i$ for any data set x, we check that $\frac{p_\theta(X)}{p_\theta(Y)} = [\theta/(1-\theta)]^{S_1(X)+S_2(X)-S_1(Y)-S_2(Y)} = 1$. So $S(X) \neq S(Y)$ doesn't imply that the ratio of likelihoods depends on θ. Thus S is not minimal sufficient, by the working definition.

☕ **(Your turn.)** Can you reproduce the arguments above to show that for the Poisson-generated data with T and S defined earlier, T is minimal sufficient, but not S?

tl;dr...

If you didn't want / have time to read the whole story above, at least take home these morals.

1. Sufficiency is a criterion for knowing how much we can afford to use compressed versions of original data without losing information, or the ability to perform maximum likelihood estimation. (See Figure 1 below.)
2. To check if a summary T of the data $\{X_i\}_{i=1}^n$ is sufficient, use the factorisation theorem (Theorem 2) by constructing g_θ and h and verifying eq. (1).
3. To check if a summary T of the data is minimal sufficient, use the minimal sufficiency criterion (Theorem 4) by verifying both directions of implication in (3).
   - The $\Leftarrow$ is always true if T is sufficient, but T needs to be minimal sufficient for $\Rightarrow$ to be true.
   - It helps to verify the contrapositive for $\Rightarrow$. (See worked example above.)

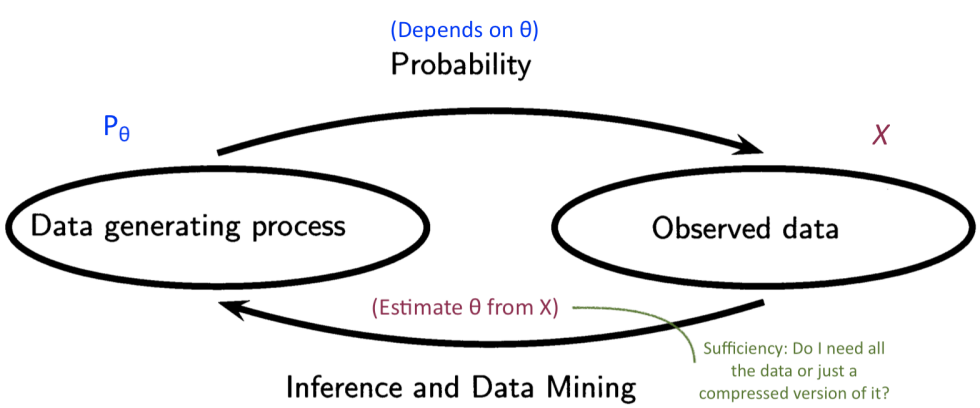

Figure 1: When estimating θ from the data set X, can I afford to store and use a smaller-dimensional summary of X (to save storage space)? (Figure adapted from Wasserman's *All of Statistics*.)